

AHANAF MOKAMMEL OMI

Full-Stack Developer

Dhaka-1205, Bangladesh | ahanafmo@gmail.com | [Portfolio](#)
github.com/BigSmoke4 | linkedin.com/in/ahanaf-mokammel-omi

Summary

Full-Stack Developer experienced in building scalable web applications and backend systems with ASP.NET Core, C#, SQL Server, PostgreSQL, Redis, and JavaScript. Strong in system design, database optimization, distributed systems, and AI-powered applications, with a focus on clean code, performance, and reliable software delivery.

Experience

Junior Web Developer – Billing Master Software Limited, Bangladesh.

2025 – Present

- Built 3+ ASP.NET MVC modules, improving internal processing efficiency by 30%.
- Implemented scalable CRUD and database features that reduced user-reported errors by 25%.
- Optimized SQL queries resulting in 20% faster data retrieval.

Executive Member – BRAC University Computer Club, Bangladesh.

2022 – 2025

- Organized university-wide coding workshops and hackathons with 100+ student participants.
- Collaborated with the web team to maintain club website and event registration systems.

General Member – Dhaka Residential Model College Computer Club, Bangladesh

2018 – 2020

- Participated in intra-school programming competitions and basic web development training sessions.

Projects

NEXUS – Supply Chain Intelligence Command Center

[GitHub](#)

Technologies: ASP.NET Core MVC, C#, PostgreSQL, Redis, EF Core, Docker

- Engineered an enterprise supply-chain decision-support platform modeling dependencies as graphs; implemented BFS/DFS traversal, alternate-path discovery, single-point-of-failure detection, bottleneck analysis, and downstream impact propagation.
- Built deterministic day-by-day disruption simulation with inventory depletion, stockout-day, service-level, recovery-time, revenue-at-risk, multi-scenario what-if analysis, AI specialist orchestration, RBAC, audit logging, SignalR, and Testcontainers.

VERITAS – Enterprise Identity & Access Governance Platform

[GitHub](#)

Technologies: ASP.NET Core, C#, PostgreSQL, Redis, EF Core, Identity

- Built an enterprise IAM and zero-trust authorization platform supporting RBAC, ABAC, deny-by-default policy evaluation, ALLOW/DENY/APPROVAL outcomes, immutable policy versions, risk scoring, and privileged access controls.
- Implemented risk-aware MFA workflows using IP, login failure, privileged action, after-hours, and device-trust signals; added JIT access, access reviews, API-key hashing/rotation, and HMAC-SHA256 webhooks.

NEURA – Neural AI Agent Orchestration & Continuity Platform

[GitHub](#)

Technologies: ASP.NET Core, C#, PostgreSQL, Redis, SignalR, OpenAI, Anthropic, Google AI

- Developed a multi-agent orchestration platform maintaining task continuity across AI providers through context monitoring, ContextHandoffPackage generation, validation, automatic handoff, weighted agent routing, and provider abstraction.
- Implemented memory, knowledge graphs, real AI adapters, retry/backoff, circuit breakers, rate limiting, encrypted credentials, cost tracking, and real-time SignalR/Cytoscape mission visualization.

FORGE – AI Software Engineering Intelligence Platform

[GitHub](#)

Technologies: ASP.NET Core, C#, Roslyn, PostgreSQL, EF Core, Git

- Engineered a software-engineering intelligence pipeline combining Roslyn semantic analysis, repository indexing, C# symbol extraction, dependency graphs, Git-history mining, and deterministic security/technical-debt analysis.
- Implemented deterministic software-risk scoring, security pattern scanning, namespace architecture policies, context retrieval, multi-tenant authZ, audit logging, rate limiting, and modular-boundary enforcement.

PULSE – Distributed Job Processing & Workflow Platform

[GitHub](#)

Technologies: ASP.NET Core MVC, C#, RabbitMQ, Redis, SQL Server, EF Core

- Engineered a distributed job-processing modular monolith using RabbitMQ durable priority queues, manual acknowledgements, dead-letter queues, Redis distributed locks, idempotency protection, and worker heartbeats.
- Implemented job lifecycle management, failure/retry transitions, exponential backoff, cancellation, scheduled jobs, worker orchestration, queue monitoring, and failure simulation.

BDPRICEFINDER – Multi-Vendor E-Commerce Platform

[Website](#)

Technologies: ASP.NET MVC, Razor Views, C#, HTML, CSS, JavaScript, SQL Server

- Developed a full-stack Bangladesh multi-vendor price-comparison marketplace supporting buyers, sellers, and vendors through role-specific dashboards, product/vendor workflows, approval processes, and hierarchical permissions.
- Implemented product lifecycle management, dynamic variants, inventory/order management, 2-step email OTP auth, coupons, EMI, warranty, Elasticsearch search, Redis caching, and vendor-aware marketplace functionality.

LOCAL AGENT – Local-First Autonomous Coding Platform

[GitHub](#)

Technologies: ASP.NET Core, C#, PostgreSQL, EF Core, Ollama, Roslyn, SignalR

- Built a local-first autonomous software-engineering agent platform with persistent sessions, model-driven DAG task planning, dependency validation, fan-out/fan-in execution, retry budgets, and verification loops.
- Implemented SHA-256 incremental repository indexing, Roslyn symbol extraction, cross-file reference graphs, context selection, and layered short/long-term memory with semantic and lexical retrieval.

ATLAS – Enterprise Intelligent API & Event Platform	GitHub
<i>Technologies: ASP.NET Core, C#, PostgreSQL, Redis, Kafka, OpenTelemetry</i>	
- Engineered a modular enterprise platform combining API management, service registry, traffic routing, Kafka event processing, retries, DLQ/replay, Redis Lua rate limiting, circuit breakers, and tenant authorization. - Implemented OpenTelemetry-based SLO/error-budget analysis, health probing, incident state management, deployment regression analysis, and audit/security controls.	
AURORA – Organizational Resilience & Decision Intelligence Architecture	GitHub
<i>Technologies: ASP.NET Core MVC, C#, PostgreSQL, EF Core, SignalR</i>	
- Developed an enterprise digital-twin platform modeling organizational assets through graph intelligence; implemented BFS/DFS, Dijkstra, upstream/downstream propagation, SPOF, and bottleneck scoring with disruption simulation. - Added weighted risk scoring, AI investigation, human-in-the-loop approvals, audit trails, multi-tenancy, and SignalR visualization while explicitly separating implemented functionality from planned RAG and multi-agent features.	
SENTINEL – Enterprise Operational Risk & Resilience Simulator	GitHub
<i>Technologies: ASP.NET Core MVC, C#, PostgreSQL, Redis</i>	
- Built an operational-risk simulation platform using dependency graphs, BFS/DFS traversal, single-point-of-failure detection, critical-path analysis, weighted failure propagation, and business-impact modeling. - Implemented genuine Monte Carlo simulation using normal/log-normal sampling, recovery-strategy weighted scoring, resilience metrics, risk heatmaps, background jobs, and interactive graph visualization.	
AEGIS – Intelligent Incident Response & Digital Operations Platform	GitHub
<i>Technologies: ASP.NET Core MVC, C#, PostgreSQL, Redis</i>	
- Developed an SRE/AIOps platform with a synthetic infrastructure digital twin implementing the Failure → Detection → Correlation → Root Cause → Impact → Remediation pipeline. - Implemented rolling z-score anomaly detection, graph-based incident correlation, weighted root-cause ranking, synthetic business-impact analysis, and human-approved remediation runbooks.	
NIRBHOR – Bilingual AI Personal Administrative Agent	GitHub
<i>Technologies: ASP.NET Core MVC, C#, PostgreSQL, EF Core, Identity, SignalR, Anthropic</i>	
- Built a bilingual AI administrative agent supporting English, Bangla, and mixed-language workflows for tasks such as NID correction and passport renewal through conversation, language understanding, and tool execution. - Implemented versioned prompt management, multilingual evaluation datasets, accuracy-gated evaluation, audit logging, notifications, cost tracking, OpenTelemetry, and secure Identity workflows.	
E-COMMERCE – ASP.NET MVC/Razor Web Application	GitHub
<i>Technologies: ASP.NET MVC, C#, Razor Views, Entity Framework, SQL Server</i>	
Developed a responsive database-driven e-commerce application using ASP.NET MVC and Razor with separated Controllers, Models, Services, and Views, supporting product management and shopping-cart functionality.	

Skills	
Programming Languages: C#, Python, JavaScript.	
Web Technologies: ASP.NET, Razor Views, HTML5, CSS3, FastAPI, REST APIs.	
Databases: SQL Server, PostgreSQL, MySQL, Entity Framework Core.	
Distributed Systems & Infrastructure: Redis, Docker.	
Software Architecture: Modular Monolith, Layered Architecture, Domain-Driven Design concepts, Application Services, Dependency Injection, RESTful APIs.	
Algorithms & Engineering: BFS, DFS, Dependency Graphs, Failure Propagation, Monte Carlo Simulation, Risk Scoring, Root-Cause Analysis, Optimization, Caching, Query Optimization.	
AI & Machine Learning: YOLO, Vision Transformers (ViT), DINOv2, Swin V2, CNN, Random Forest, HOG + SVM, Supervised Contrastive Learning + Classification.	
Testing & Tools: k6, GitHub, Visual Studio, Google Data Studio, Retool, Microsoft 365.	
Certificates	
Foundational C# Developer – Microsoft & freeCodeCamp (2026)	Understand ASP.NET Core Fundamentals – Microsoft (2026)
CSS Essentials – Cisco Networking Academy (2026)	Python Essentials 2 – Cisco Networking Academy (2026)
JavaScript Essentials 1 – Cisco Networking Academy (2026)	English for IT 2 – Cisco Academy B2 Prof. (2026)
Legacy Responsive Web Design V8 – freeCodeCamp (2026)	Software Engineer Role – HackerRank (2026)
Google Analytics Certification – Google (2026)	HubSpot SEO II Certified – HubSpot Academy (2026)
Education	
B.Sc. in Computer Science – BRAC University, Bangladesh	2026
Higher Secondary Certificate – Dhaka Residential Model College, Bangladesh	2020
Secondary School Certificate – Ibn Taimiya School and College, Bangladesh	2018
Languages	
English – Professional Working Proficiency Bengali – Native Speaker	